Vasanth Pidaparthy

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PROFESSIONAL SUMMARY

PhD candidate in mathematics (expected graduation MAY 2027) working on geometric analysis and nonlinear PDE. Experience formulating applied problems geometrically and solving them numerically in JAX. Looking for research roles at the interface of mathematics and AI.

EDUCATION

University of Maryland, College Park, MD
PhD in Mathematics | AUG 2021 – MAY 2027 (expected) | GPA: 3.9/4.0
Advisor: Prof. Yanir A Rubinstein | Thesis: Weak and strong solutions to non-linear elliptic PDE

Indian Institute of Science, Bengaluru
– Master of Science in Mathematics | AUG 2020 – MAY 2021| First class w/ distinction | GPA: 9.9/10
– Bachelor of Science (Research) in Mathematics | AUG 2016 – MAY 2020 | First class w/ distinction | GPA: 9.5/10

INTERNSHIP EXPERIENCE

Quantum Computing Research Intern, Error Corp. | JUN 2025 — AUG 2025 | 10 weeks (Full time)
– Modelled the problem of minimizing first order interaction/cross-talk errors in a 2-qubit quantum system as a differential geometry problem of identifying optimal control curves in the lie group SU(2).
– Implemented a numerical solver in Python (JAX) to optimize control pulses, achieving a significant reduction in error rates. Delivered a Python repository.

RESEARCH SUMMARY

Subequations and Regularity: Optimal regularity up to the boundary for solutions to the geometric PDE defined by a subequation, with applications to Monge–Ampère and special Lagrangian equations.

Interacting LLM Agents: Studying collective behavior in social networks of LLM-based agents and developing theoretical and mathematical approaches to characterize their dynamics.

DSL: Regularity up to the boundary fails in all branches of the degenerate special Lagrangian (DSL) equation under Lagrangian boundary conditions. Weak solutions exist for time-convex boundary conditions in the top two branches, and are (i) convex in the top branch, and (ii) time-convex in the second branch.

Special Lagrangians with boundary: Deformations of special Lagrangians with Lagrangian boundary conditions are unobstructed, and the corresponding moduli space admits a pair of dual special affine structures constructed using Lagrangian flux functionals.

TALKS

(upcoming) **40th Geometry Festival** | UMD, College Park, Maryland, USA | APR 2–4, 2027
Geometry & Topology Seminar | IISc, Bengaluru, India | SEP 29, 2025

BOOKS

Subequations | Graduate textbook | Accepted in AMS Graduate Studies in Mathematics (subject to revisions) V. Pidaparthy, Y.A. Rubinstein | 315 pages

User's guide to the special Lagrangian equation | Graduate textbook (in preparation) | Solicited by the American Mathematical Society (AMS) | V. Pidaparthy, Y.A. Rubinstein

PUBLICATIONS

Convexity and the degenerate special Lagrangian equation | July 2025 | arXiv:2507.17018 V. Pidaparthy, Y.A. Rubinstein

Moduli of special Lagrangians with boundary, II: Lagrangian Flux and Affine Structures | Mar 2025 | arXiv:2503.23532 | V. Pidaparthy

Moduli of special Lagrangians with boundary, I: Unobstructed Deformations | to appear in Indiana University Mathematics Journal |arXiv:2503.23525 | V. Pidaparthy

The Hausdorff distance and metrics on toric singularity types | Bulletin des Sciences Mathématiques | Sep 2025 |A. Aitokhuehi, B. Braiman, D. Cutler, T. Darvas, R. Deaton, P. Gupta, J. Horsley, V. Pidaparthy, J. Tang

SKILLS

Technical Skills: Python (NumPy, JAX, Pandas), LaTeX

Relevant Coursework: probability, measure theory, Riemannian geometry, information theory, geometric analysis, algebraic geometry, Lie groups, combinatorics, algorithms and data structures

Mathematical Modeling: PDEs, differential geometry

Professional Skills: flexibility in working on diverse research projects, collaborative problem-solving, communication of complex mathematical ideas, mentorship experience guiding students and peers

AWARDS AND HONORS

Patrick and Marguerite Sung Fellowship in Mathematics | UMD | AUG 2025 – DEC 2025

Hauptman Summer Fellowship | UMD | MAY 2024 and MAY 2022

CMNS Outstanding Teaching Assistant Award | UMD | MAY 2022

Dean's Fellowship | UMD | AUG 2021 – MAY 2023

Kishore Vaigyanik Protsahan Yojana (KVPY) Scholarship | India | AUG 2016 – MAY 2021

LEADERSHIP AND TEACHING

Graduate Teaching Assistant, Department of Mathematics, University of Maryland, College Park
AUG 2021 — PRESENT | **Responsibilities:** Classroom instruction, designing and grading quizzes and exams.

Graduate Teaching Assistant, Research Experience for Undergraduates (REU)
– JUN 2024 — JUL 2024 | Convex and Complex Geometry | 6 undergrads | Publication (Bull. Sci. Math., 2025)
– JUN 2023 — JUL 2023 | Geometric Measure Theory | 4 undergrads

Co-organizer, Student Complex Geometry Seminar, UMD | AUG 2023 – MAY 2024
Delivered a semester-long lecture series on Gromov–Hausdorff convergence of Ricci lower bounded manifolds.

Social Coordinator, Graduate Student Council of MATH, AMSC, and STAT | AUG 2023 – MAY 2024
Created and organized the JAMS student seminar to give grad students exposure to broad fields of mathematics

Mentor, Directed Reading Program (DRP) | FEB 2023 – MAY 2023 | Topic: The Bolzano–Weierstrass theorem

Instructor, Monsoon Math Camp | JUN 2021 | Online high school math camp in India | Probability, martingales, and gambler's ruin